

Numbers and Their Types - Maths Notes

What are Numbers?

Numbers are symbols used to count, measure, compare, and solve mathematical problems. Examples: 1, 5, 100, -3, $\frac{1}{2}$, 3.14

Natural Numbers

Counting numbers starting from 1.

Examples: 1, 2, 3, 4, 5

Whole Numbers

Natural numbers including zero.

Examples: 0, 1, 2, 3, 4

Integers

Positive numbers, negative numbers, and zero.

Examples: -3, -2, -1, 0, 1, 2

Rational Numbers

Numbers that can be written in fraction form $\frac{p}{q}$ where $q \neq 0$.

Examples: $\frac{1}{2}$, $\frac{3}{4}$, $-\frac{5}{7}$

Irrational Numbers

Numbers that cannot be written as fractions.

Examples: $\sqrt{2}$, π

Real Numbers

Combination of rational and irrational numbers.

Even Numbers

Numbers divisible by 2.

Examples: 2, 4, 6, 8

Odd Numbers

Numbers not divisible by 2.

Examples: 1, 3, 5, 7

Prime Numbers

Numbers having exactly 2 factors: 1 and itself.

Examples: 2, 3, 5, 7

Composite Numbers

Numbers having more than 2 factors.

Examples: 4, 6, 8, 9

Positive Numbers

Numbers greater than zero.

Examples: 1, 2, 10

Negative Numbers

Numbers less than zero.

Examples: -1, -2, -10

Quick Summary Table

Type	Examples	Fractions?	Negatives?
Natural	1,2,3	No	No
Whole	0,1,2	No	No
Integers	-2,0,5	No	Yes
Rational	$\frac{1}{2}$, 3	Yes	Yes
Irrational	$\sqrt{2}$, π	No	Sometimes
Real	All normal numbers	Yes	Yes

Important:

Integers, Fractions, Rational Numbers, and Real Numbers are the most important concepts for higher mathematics, algebra, aptitude, and programming logic.